

SECOND ORDER

AI, Capital & Work · A Second Order research briefing

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The First Job Moved

Young workers in the most exposed occupations are 19 percent behind, and the number is true. The employed share of the age group is unchanged, their pay is rising fastest, and the firms and occupations they are entering are not the ones that collected the toll.

An account of one number and the record around it, followed from primary data and held open to correction.

FOR DISCUSSION · NOT INVESTMENT ADVICE

Is the machine taking the first job?

THE THROUGH-LINE

The first job in knowledge work was a toll: years of documented, checkable work paid inside one firm's hierarchy for the judgment that comes only from doing the work. A machine now does the documented part, so the incumbents that collected the toll collect less of it, and the young are entering by other routes, at the firms that adopted, the firms they are founding, and the occupations that did not exist two years ago.

WHY NOW

The Stanford figure widened from 15 to 19 percent in August, the Fed chair named recent graduates as the one labor concern at Jackson Hole, and the August employment report on September 4 will be treated as the verdict. The household survey, the firm data and the formation statistics already say what the verdict cannot.

WHAT YOU'LL LEARN

1. The number is correct at its grain, and the grain is the worker inside an exposed occupation (01)
2. The employed share of 20-to-24-year-olds is unchanged, and their pay growth leads (02)
3. The tool compresses the distance from beginner to competent by about three quarters (03)
4. Retirements will open twelve times as many jobs a year as growth through 2035 (04)
5. The findings align once they are placed by grain (05)

IF YOU REMEMBER ONE LINE: the machine is collecting less of the toll the first job used to charge, and the open question is whether the shortcut skips the learning.

Where the young are entering knowledge work

Workers aged 22 to 25 in the most exposed occupations are 19 percent below the same age group in the least exposed. The share of all 20-to-24-year-olds in work was 65.4 percent in mid-2026 and in mid-2025 alike, their pay is rising faster than any other age group's, and the heaviest AI adopters added entry-level staff 12 percent faster than firms that had not yet adopted.

Second Order · Issue 11 claim

19% vs +10%

employment of 22-to-25-year-olds in the most exposed occupations against the least exposed, November 2022 to June 2026 (Stanford, ADP records).

65.4%

the share of 20-to-24-year-olds in work in the three months to July 2026, the same as a year earlier; unemployment and participation fell together (BLS).

12x

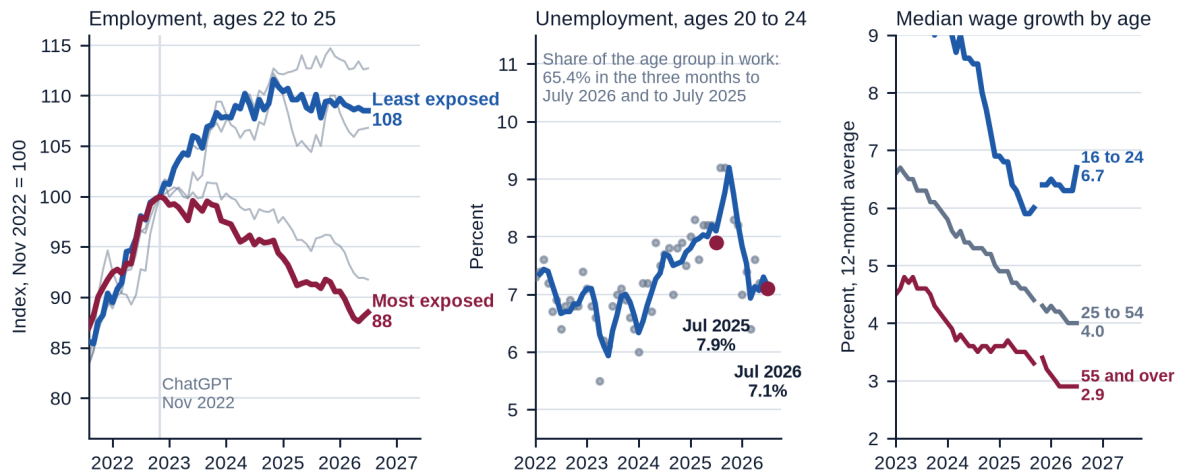
the ratio of annual job openings from workers leaving the labor force to openings from growth, 2025 to 2035 (BLS projections, August 27).

The most exposed door closed on the young; the age group did not vanish

EXHIBIT 1

The most exposed door closed on the young, and the age group did not vanish

Employment of 22-to-25-year-olds by AI exposure (Nov 2022 = 100); unemployment rate, 20 to 24; median wage growth by age



Note: Exposure quintiles are the Stanford authors' construction. Unemployment: monthly points and three-month average; no October 2025 survey.

Source: Stanford Digital Economy Lab, Canaries dashboard (ADP records); BLS CPS via FRED; Atlanta Fed Wage Growth Tracker; to July 2026.

SO WHAT

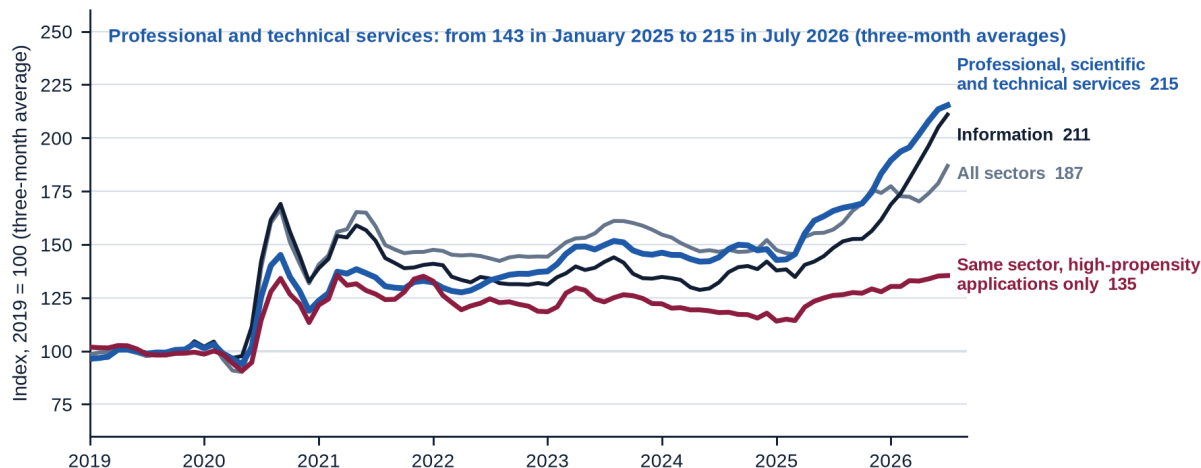
- Employment of 22-to-25-year-olds in the most exposed fifth of occupations fell to 88 by June 2026 while the least exposed rose to 108 (November 2022 = 100), on Stanford's assembly of ADP payroll records.
- The gap arose through hiring, with separations unchanged; workers over 30 in the same occupations grew; where the machine assists the work without doing it, the young held up.
- The Yale Budget Lab finds no change in the occupational mix, which is consistent with a recomposition inside occupations and across firms.

Applications to form professional firms run at twice the 2019 rate

EXHIBIT 2

Applications to form professional and technical firms run at twice the 2019 rate

Business applications, seasonally adjusted, indexed to the 2019 monthly average = 100, January 2019 to July 2026



Note: Applications for an employer identification number, not firms. High-propensity applications show traits of future employers.

Source: U.S. Census Bureau, Business Formation Statistics, monthly release of August 12, 2026; Second Order index.

SO WHAT

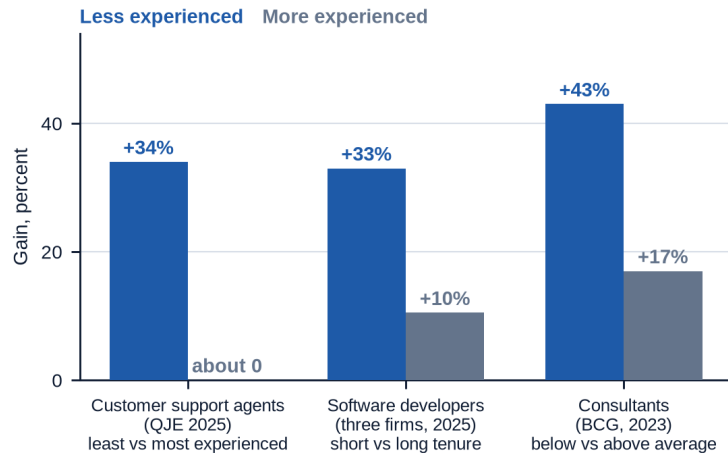
- Unemployment among 20-to-24-year-olds was 7.1 percent in July 2026 against 7.9 a year earlier, and participation fell by as much, so the employed share of the age group, 65.4 percent, was unchanged.
- Firms in the top third of AI spending were 10 percent larger two years after adoption with entry-level headcount 12 percent larger (Ramp and Revelio, 21,559 firms; adopters were already growing faster).
- Applications to form professional and technical firms rose from 143 to 215 on the 2019 index between January 2025 and July 2026; the stricter high-propensity measure rose 5 percent.

In every controlled study the least experienced gain most from the tool

EXHIBIT 3

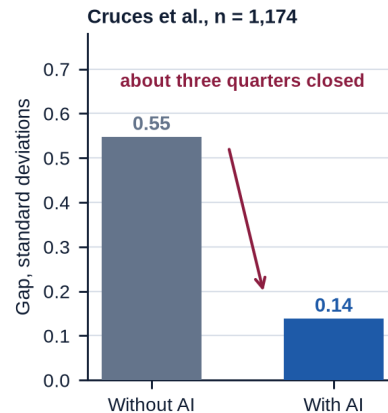
In every controlled study the least experienced gain most from the tool

Left: productivity gain with an AI assistant, percent, by experience or ability. Right: the education gap in output with and without AI



Note: Developer figures are midpoints of reported ranges (27 to 39 and 8 to 13 percent). Four settings, three metrics; not pooled.

Source: Brynjolfsson, Li and Raymond, QJE 2025; Cui et al. 2025 (three firms); Dell'Acqua et al., HBS 2023; Cruces et al., arXiv, August 2026.



SO WHAT

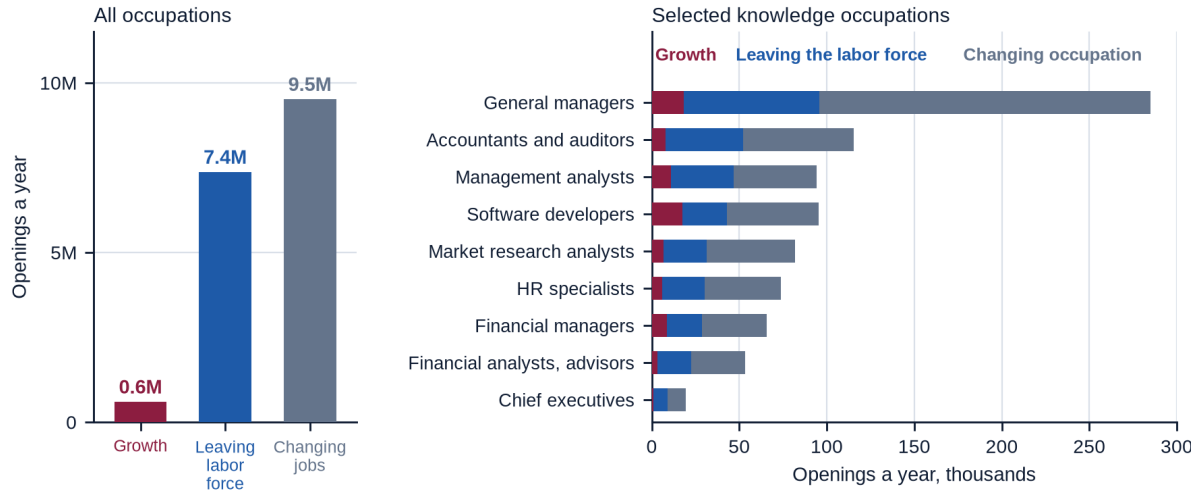
- Customer-support agents gained 34 percent for the least experienced against about zero for the most (QJE 2025); developers 27 to 39 percent for short tenure against 8 to 13 (three firms, 2025); consultants 43 against 17 (BCG, 2023).
- Cruces and colleagues found the education gap in output fell from 0.55 to 0.14 standard deviations with the tool, about three quarters of the way to zero (August 2026).
- Four settings, three metrics, no field replication in law, consulting or medicine after 2024; the finding is the agreement across the four.

Retirements open twelve times as many jobs a year as growth through 2035

EXHIBIT 4

Through 2035, retirements open twelve times as many jobs a year as growth does

Projected annual openings by source, 2025 to 2035: all occupations (millions) and selected knowledge occupations (thousands)



Note: Growth is the ten-year employment change divided by ten. Legal, healthcare and trades rows were not retrievable this pass.

Source: Bureau of Labor Statistics, Employment Projections 2025-35, Table 1.10 (August 27, 2026).

SO WHAT

- BLS projects 17.5 million openings a year to 2035: 7.4 million from workers leaving the labor force, 9.5 million from workers changing occupation, about 590,000 from growth.
- Among accountants and auditors, 44,100 a year will leave against 7,900 added by growth; among software developers, 25,600 against 17,500.
- The old occupations are farmers, judges, executives, accountants and physicians, where judgment is the product; electricians and engineers are younger than the workforce.

The two camps measure different cells of one recomposition

EXHIBIT 5
The two camps measure different cells of one recomposition

Published findings placed by the grain at which each was measured (rows) and by what it measures (columns)

	quantity	price	quality
occupation mix	● Occupation mix, unemployment by exposure unchanged (Yale)		
industry	● Adopting industries post no fewer jobs (Fed Board)		
occupation	● Exposed occupations +25 pct jobs, -2.3 pct real wages (Second Order)	● Wage growth 6.7 points slower in exposed occupations (Apollo)	
firm	● Heavy adopters: entry-level headcount +12 pct (Ramp, Revelio)		
firm formation	● Professional-services applications 2.2x 2019 (Census)		
worker type	● Job-finding down 13 points for attached workers (Richmond Fed)		
worker by occupation	● 22 to 25 in most exposed jobs, 19 pct below benchmark (Stanford)		
age group	● Unemployment 20 to 24: 7.1 pct vs 7.9 a year ago (BLS)	● Under-25 wage growth 6.7 pct, fastest of any age (Atlanta Fed)	
posting	● Freelance posts -21 pct; surviving posts pay more (Demirci)		
task	● Software postings +15 pct since Feb 2025, 71 pct senior (Indeed)		● Learning with AI: 50 vs 67 pct on test, no time saved (Anthropic)
	● Novices gain most: +34 vs 0, +33 vs +10, +43 vs +17 (three trials)		
	● rises	● falls	● no change ● mixed

Note: Direction is the sign each source reports at its own grain. Mixed marks a finding with quantity up and price down.
Source: Sources as named in each cell; Second Order arrangement.

SO WHAT

- At the occupation mix, nothing; at the firm, entry-level hiring up; at the worker inside an exposed occupation, 19 percent behind; at the age group, employed share unchanged and pay growth leading; at the task, novices gain most.
- Apollo finds real wage growth 6.7 points slower in exposed occupations, worst in the lowest-paid quartile, with no employment effect: the documented share of the work is what reprices, whoever does it.
- Carta shows Series D headcount down 29 percent from 2023 while AI staff pay rose 9 percent; the loss falls on the hierarchy built to collect the toll.

Takeaways

1. The grain decides the finding

Occupation-mix trackers, firm-level studies, worker-level payroll data and task-level experiments each report a different cell of one recomposition; a claim that cites only one of them describes a single unit of measurement.

2. The documented share is what reprices

Wage evidence from Apollo, the freelance platforms and the occupational survey shows quantity steady and price falling at the documented end of the work, with price rising at the judgment end, whoever does it.

3. The openings come from departures

Departures from the labor force will open 7.4 million jobs a year to 2035 against 590,000 from growth, heaviest in the judgment occupations, which makes faster competence the scarce good.

4. The learning is the open question

One trial of 52 developers found those who learned with an assistant scored two letter grades lower with no time saved; a second trial, larger and outside software, is the indicator that would change our mind.

What would change our mind

The August employment report, released September 4: the 20-to-24 unemployment and participation rates and employment in information and computer systems design; a negative print changes none of the grain-level findings.

Mid to late September: the Stanford dashboard, Ramp's adoption index, the Yale tracker after the August household survey, the Census Bureau's August formation statistics, and the August JOLTS report on September 29.

The Federal Open Market Committee, September 15 and 16: with new projections, its chair having named recent graduates as the one area of labor-market concern.

The Shen and Tamkin design, repeated: a second learning trial with more than 52 subjects and a task outside software; if the assisted group learns as much, the cost in the second-order box disappears.

The falsifier: the employed share of 20-to-24-year-olds falling with participation, high-propensity formation in professional services turning down while total applications rise, or field replications showing the gains concentrated among experts.

IN ONE LINE. The first job was a toll; the toll is being collected less and falling at once; what remains to be tested is whether the shortcut skips the learning the toll used to provide.